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PRINCIPLES OF MANAGEMENT

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Module 2

Module II

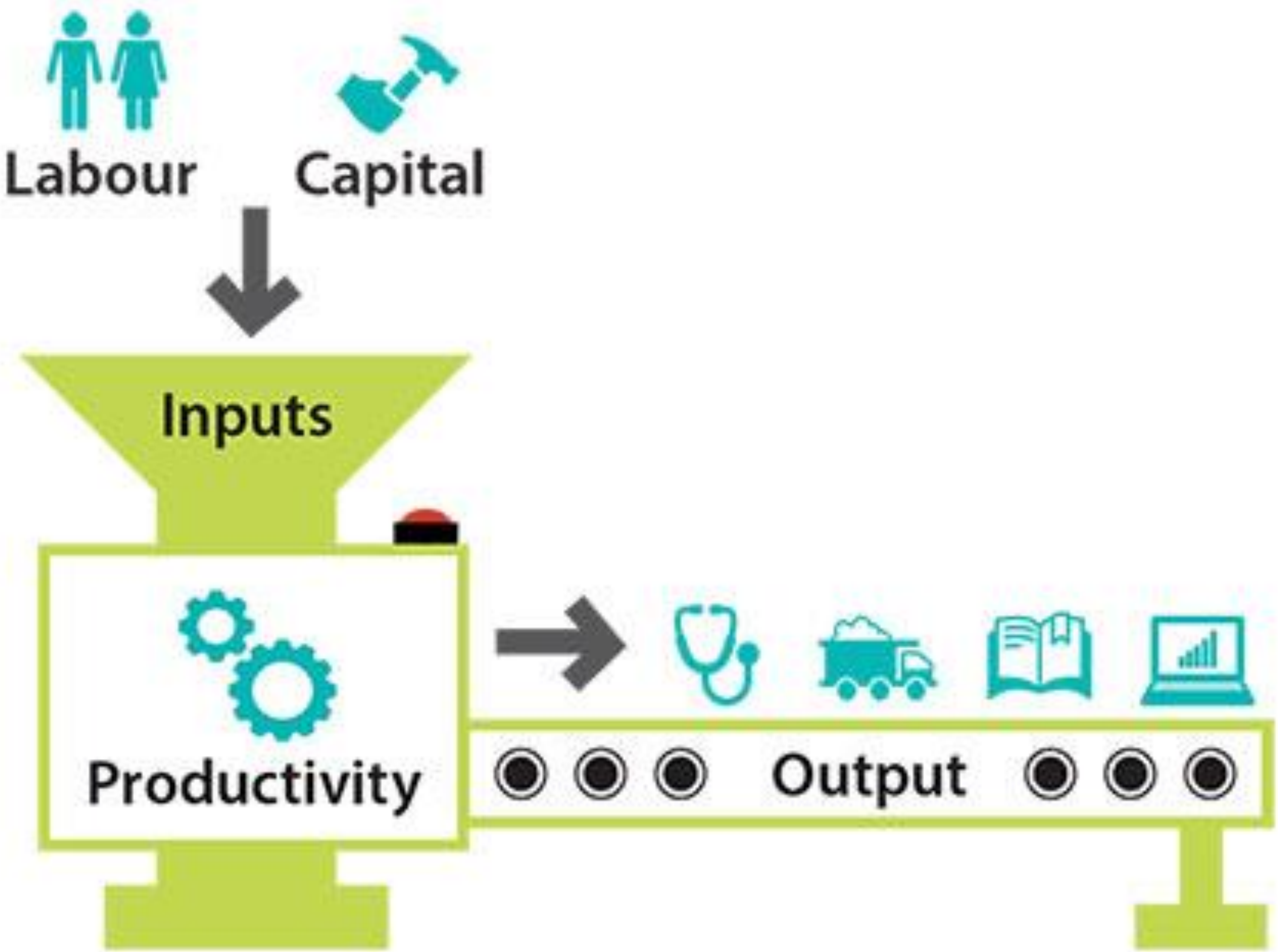
Production planning and control: Objectives and functions.

Production management: Structure, objectives, productivity index, modern productivity improvement techniques.

Inventory Management: Functions, classifications of inventory, basic inventory models, inventory costs, Economic order quantity (EOQ). Materials Requirement Planning – Objectives, Functions and methods.

Project Management: Functions, Characteristics, Feasibility studies, Project network analysis – PERT/CPM.

Productivity and Production



Productivity and Production

- Productivity is a measure of how much input is required to produce a given output
  - **Productivity = output/input**
  - **Productivity = Value of goods & services produced / Value of resources utilized for this production.**
- Productivity is the measure of how well the resources are brought together in an organization and utilized for accomplishing a given set of objectives.
- It can be defined as the output – input ratio within a time period with due consideration for quality.
- It is applicable to all production systems

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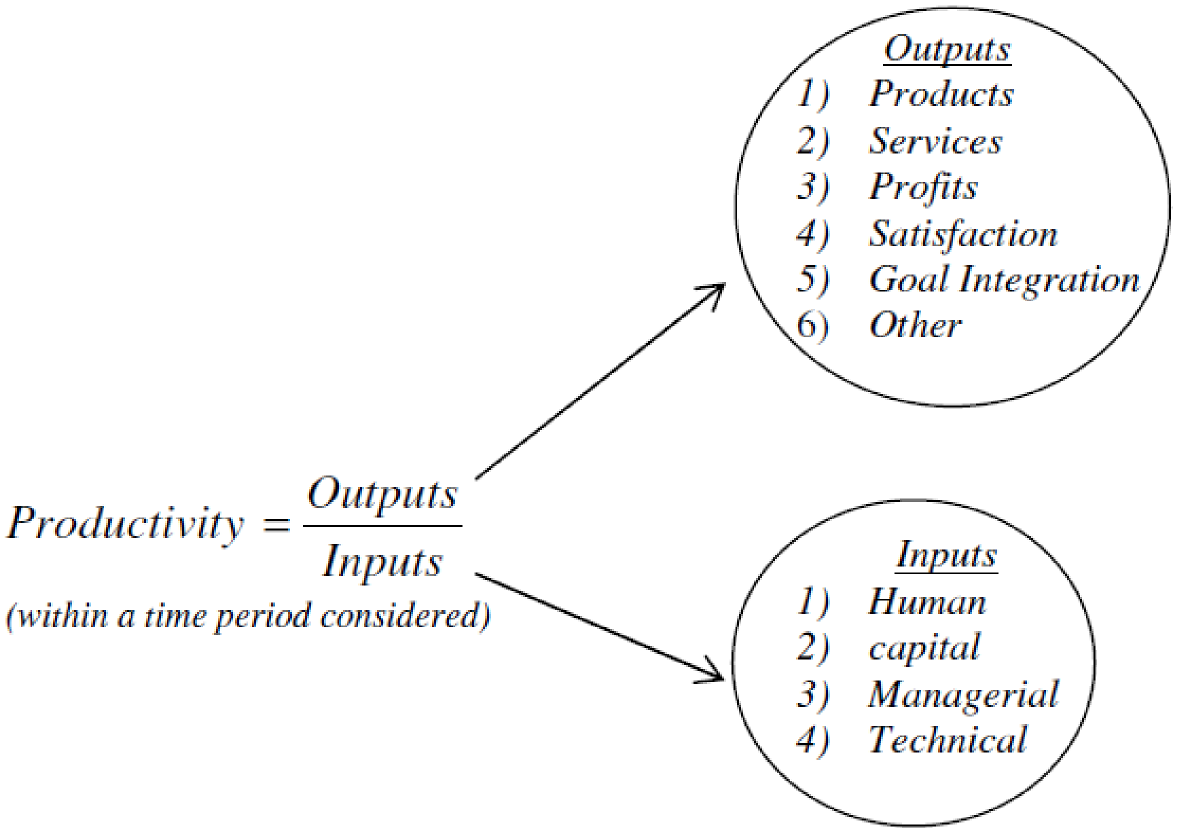
Productivity and Production



Productivity and Production

- The formula indicates that productivity can be improved by:
    - Increasing the outputs with the same inputs
    - Decreasing the inputs by maintaining the same outputs or
    - Increasing the outputs and decreasing the inputs to change the ratio favourably.
- Production and productivity are different terms and implies different meaning. It should be noted that higher production need not necessarily lead to higher productivity and vice versa.

Productivity

- Productivity implies effectiveness and efficiency in individual and organizational performance.
- Effectiveness is the achievement of objectives whereas efficiency is the achievement of ends with the least amount of resources.
- Managers cannot know whether they are productive unless they first know their objectives and goals and those of the organization.

Production

- Production is a process (or system) of converting input into some useful, value added output.

- Production is a measure of output produced. The emphasis is not on how well the input-resources are utilized.

- Productivity, on the other hand, puts emphasis on the ratio of output produced to the input used. That is, here the focus is on how well the input resource is used for conversion into output.

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Productivity Vs Production

Productivity	Production
• Relative term • A measure of efficiency of system • Increases by efficient utilization of inputs (4 M's *) • It may or may not have a unit	• Absolute term • A measure of capacity of system • Increases by utilizing more inputs. • It is expressed in terms of a unit (No of units or monetary terms)

** Money, material, machine* and *manpower* are the 4 M's*

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Problem

A company is manufacturing 24,000 components per month by employing 100 workers in 8 hours shift. Calculate the productivity?

Answer

Output in terms of production $= 24000$ components

Input in terms of man-hours $= 100\text{worker s} \times 8\text{hours} \times 30\text{days}$

(assuming 30 days in a month)

$= 24{,}000$ man-hours

$$\text{Productivity} = \frac{24{,}000 \text{ components}}{24{,}000 - \text{man hours}}$$

$= 1$ component/man hour

Problem

- *Suppose the company gets additional order to supply 6000 more components and the management decides to employ additional workers, calculate the productivity when the number of additional workers employed are* ***(i) 30 (ii) 25 and (iii) 20.***

Answers

(i) When additional 30 workers are employed

$$\text{Productivity} = \frac{24{,}000 + 6{,}000}{(100 + 30)(8)(30)}$$

$$= 0.96 \frac{component}{man - hour}$$

Problem

(ii) When additional 25 workers are employed

$$\text{Productivity} = \frac{24{,}000 + 6{,}000}{(100 + 25)(8)(30)}$$

$$= 1\,\frac{\textit{component}}{\textit{man} - \textit{hour}}$$

(iii) When additional 20 workers are employed

$$\text{Productivity} = \frac{24{,}000 + 6{,}000}{(100 + 20)(8)(30)}$$

$$= 1.04\,\frac{\textit{component}}{\textit{man} - \textit{hour}}$$

Note:-

1. *Increase in production does not necessarily mean increase in productivity*
2. *Productivity is always associated with the context in which it is calculated. For example, we have calculated labour productivity*

Measurement of Productivity

- **Total productivity**
  - It's the ratio of aggregate out put to aggregate input
- **Partial productivity**
  - It's the ratio of the aggregate output to any single input
    - ❑ Land and building
    - ❑ Materials
    - ❑ Machines
    - ❑ Man power

Objectives of measurement of productivity

- To study performance of a system over time.
- To have relative comparison of different systems for a given level.
- To compare the actual productivity of the system with its planned productivity.

Calculation of productivity

- **Total Productiavity Measure (TPM) –** Total outputs to the sum of all tangible input factors (labour, materials, capital, energy, other expenses, etc.)

Total Productivity Measure (TPM) = Total outputs / Sum of all Inputs

- **Partial Productivity Measure (PPM)** - Total output to one class of input.

Partial Productivity Measures

Partial Productivity	**Mathematical Representation**
Labour Productivity	Output / Labour Input
Material Productivity	Output / Material Input
Capital Productivity	Output / Capital Input
Energy Productivity	Output / Energy Input
Advertising and Media Planning Productivity	Output / Advertising and Media Planning Input
Other Expense Productivity	Output / Other Expense Input

- **Total Factor Productivity (TFP) -** total output to the sum of associated labor and capital inputs.

Total Factor Productivity (TFP) = Total Output / (Labour + Capital Inputs)

Advantages and Disadvantages of Total Productivity Measure

Advantages	**Disadvantages**
• *More accurate representation of the total picture of the company* • *Easily related to total costs* • *Considers all quantifiable outputs and inputs*	• *Difficulty in obtaining the data* • *Requirement of special data collection system*

Advantages and Disadvantages of Partial Productivity Measure

Advantages	**Disadvantages**
• *Easy to understand and calculate* • *A tool to pinpoint improvement*	• *Misleading if used alone* • *No consideration for overall impact*

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Productivity index

- ***Material productivity*** =number of units produced/cost of material

- This can be increased by
    - *Proper choice of design*
    - *Proper training and motivation of workers*
    - *Better material planning and control*
    - *Searching for cheaper alternate material etc..*

Productivity index

❖ ***Labour productivity*** = aggregate output / expenditure from labour

The unit may be number of units/man hours or in monetary terms

This can be increased by

- ✓ *Proper selection of design and process*
- ✓ *Proper training*
- ✓ *Motivate workers by financial and other incentives*
- ✓ *Providing facilities and chances for self development*

Productivity index

❖ ***Capital productivity*** = turn over / capital employed

This can be improved by

- ✓ *Better utilization of capital resources*
- ✓ *Careful make or buy decision*

❖ ***Machine productivity*** = output / actual machine hours utilized

This can be improved by

- ✓ *Preventive maintenance*
- ✓ *Use proper machine settings*
- ✓ *Using method study techniques*
- ✓ *Using properly skilled and trained workers etc.*

❖ ***General measure of productivity***

Aggregate productivity = output /(land+labour+material+capital+other inputs)

Factors Influencing Productivity

- **Controllable factors (Internal Factors)**
  1. Product
  2. Plant and equipment
  3. Technology
  4. Materials
  5. Human factors
  6. Work methods
  7. Management style
  8. Financial factors
- **Uncontrollable factors (External factors)**
  1. Natural Resources
  2. Government Policy
  3. Political Stability

Ways to Improve Productivity

- Productivity of any system can be improved by proper use of resources and optimum utilization of system or processes. These include the following.
- **Technology Based**
  - ✓ Acquire new technology
  - ✓ Automation in assembly
  - ✓ Modern maintenance techniques
  - ✓ Energy technology
  - ✓ Flexible Manufacturing Systems (FMS)

Ways to Improve Productivity

- **Employee Based**
  - ✓Financial and non-financial incentives at individual and group level
  - ✓Suggestion scheme
  - ✓Safe work-place
  - ✓Workers participation in management

Ways to Improve Productivity

- **Material Based**
  - ✓Material Planning and control
  - ✓Purchasing, logistics
  - ✓Material storage and retrieval
  - ✓Source selection and procurement of quality material
  - ✓Waste elimination
  - ✓Use of Automated Guided Vehicle (AGV) for material transportation

Ways to Improve Productivity

- **Process Based**
  - ✓ Methods engineering and work simplification
  - ✓ Job design, Job evaluation, Job safety
  - ✓ Human factors engineering
- **Product Based**
  - ✓ Value analysis and value engineering
  - ✓ Product diversification
  - ✓ Standardization and simplification
  - ✓ Reliability engineering
  - ✓ Promotion

Ways to Improve Productivity

- **Management Based**
  - ✓Management style
  - ✓Communication in the organization
  - ✓Work culture
  - ✓Motivation
  - ✓Promoting group activity

Productivity improvement procedure

- **Improving the existing method of plant operation**

- Use of method study and work measurement procedures
  - The steps involved in it are
    - ✓ Gather all information about the existing methods
    - ✓ Present it in form of charts and diagrams
    - ✓ Do critical examination of these data (why, when , who and how)
    - ✓ Develop improved methods

Productivity improvement procedure

- **Design of part**
  - ✓ Simplify the design
  - ✓ Standardize the materials, tools, procedures, etc ..
  - ✓ Efficient system of quality control
  - ✓ Use better and economic material

Productivity improvement procedure

- **Tolerance and specifications.**
  - ✓ Designers keep tight tolerances.
  - ✓ Lack of consideration of the cost factor, knowledge of production processes and process capability
  - ✓ Designers should interact with production engineers and cost analysis group

Productivity improvement procedure

- **Effective utilization of materials**

- The utilization of material should be increased by taking improvement measures in
  - Design stage
  - Process or operation stage

Productivity improvement procedure

- **Process of manufacture**
  - ✓ Selection of proper process
  - ✓ Use of proper tools, equipments ,parameters etc

Productivity improvement procedure

- **Set up and tools**

- Use of special tooling depends upon

  - ✓ The quantity to be produced
  - ✓ Chance of repeated orders
  - ✓ The amount of labour involved
  - ✓ Amount of capital requirements

- the productivity will improve by

  - ❖ Reduction in set up time
  - ❖ Design of tooling to use full capacity of machines

Productivity improvement procedure

- **Working conditions**
  - ✓ Lighting
  - ✓ Control of temperature
  - ✓ Adequate ventilation
  - ✓ Control sound
  - ✓ Promote orderliness, cleanliness and good house keeping
  - ✓ Proper waste disposal
  - ✓ Provide protective equipments

Productivity improvement procedure

- **Material handling**
  - ✓ Use mechanical equipments
  - ✓ Proper planning and scheduling
  - ✓ Proper maintenance of equipments and training to workers
- **Plant layout**
  - ✓ It should facilitate smooth flow of materials and men
  - ✓ It should reduce distance moved by men and materials
  - ✓ Storage places should be properly arranged to reduce searching and handling

Benefits of increasing productivity

- ❑ **For management**
- ✓ to earn good profit
- ✓ to have better utilization of resources
- ✓ to stand better in the market

- ❑ **For workers**
- ✓ higher wages
- ✓ better working conditions, improved morale
- ✓ higher standards of living
- ✓ job security and satisfaction

Benefits of increasing productivity

- ❑ **For consumers**
- ✓ better quality goods at reduced prices
- ✓ more satisfaction

- ❑ **To government**
- ✓ more revenue by taxation
- ✓ improvement in economy
- ✓ better utilization of resources
- ✓ increased per capita income
- ✓ development of the nation

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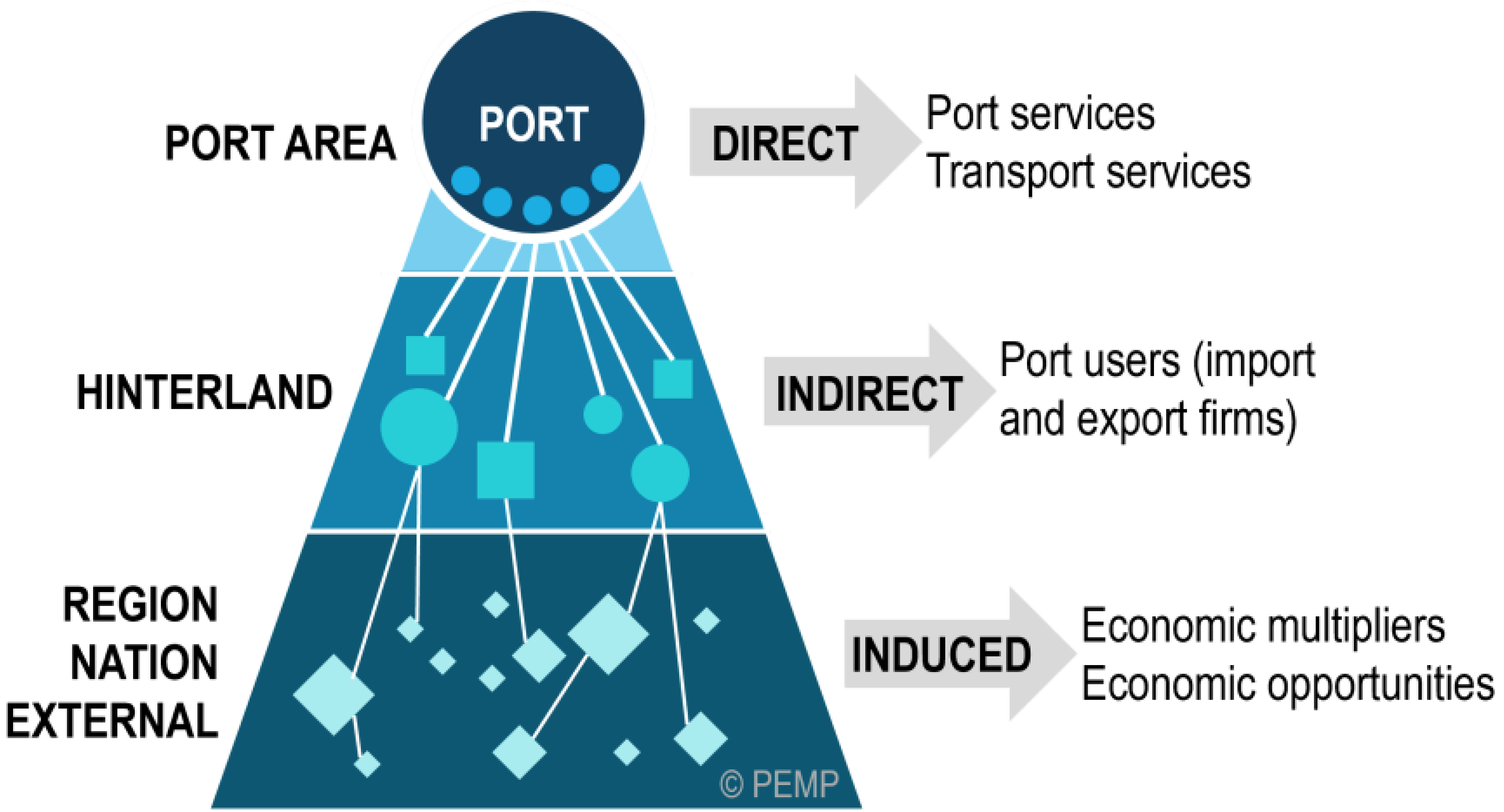
Materials Management

Material Management

- Materials are one of the **main inputs in a manufacturing plant**
- It accounts for the **50 to 85% of the total cost of production**
- Any **savings in the material leads to significant increase in profit**
- This is a basic function which **directly adds value to the product**

Material Management and Economy



[List of busiest container ports - Wikipedia]()

Material Management:- Definition

- Material management is the **planning, directing, controlling and coordinating** those activities which are concerned **with materials and inventory requirements, from the point of their inception to their introduction into the manufacturing process**
- It begins with the **determination of materials schedule** and **ends with issuance to production to meet customers demand** as **per schedule** and **at the lowest cost**
- Material management deals with **controlling and regulating the flow of material** in relation to the **changes in variables like demand, prices, availability, quality, delivery schedules etc.**

Objectives of Material Management

- **Minimizing cost** of materials
- To procure and provide materials of desired quality when required, at the lowest possible overall cost of the concern
- **Receiving and controlling material safely and in good condition**.
- **Identification of surplus stocks and taking appropriate measures to reduce it**
- To **cut down the costs through simplification**, **standardization, value analysis, import substitution etc**.
- To **train personnel in the field of materials management** in order to increase operational efficiency
- Efficient **record keeping** and prompt reporting

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Functions of Material Management

- **Planning function**
  - Translation of the sales projection into long term requirements of the materials
  - Adjust the material requirement as per updated production plan
  - Arrange the facilities required for the materials management
- **Production control**
  - Determining requirements of materials and parts to be purchased and manufactured
  - Schedules of production and purchasing
  - Issue work order to departments and purchase order to suppliers
  - To dispatch the materials to the production department

Functions of Material Management

- **Purchasing**
  - Selection of **acceptable suppliers and issue of purchase order**
  - To **expedite the delivery of materials** to **meet the inventory requirements**
  - To **look for new** materials and suppliers
- **Inventory and stores control**
  - Requisition of materials according to requirements at appropriate time
  - Keep updated records of the materials received, issued and balance
  - Do physical verification of materials

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Functions of Material Management

- **Shipping /physical verification**
  - **To receive, verify and store finished goods**
  - **Pack and label finished goods**
  - To **prepare shipping documents**
  - Report about the shipment to the accounting department and sales department
- **Material handling**
  - **To handle in-plant materials**
  - **Maintenance** of material handling equipment's

Functions of Material Management

- **Transportation/ traffic control**
  - **To carry** raw materials from the **suppliers to plant**
  - To carry **finished goods to customers**
  - To control **routes and rates of hired vehicles**
  - To **maintain the fleet of delivery vehicles**

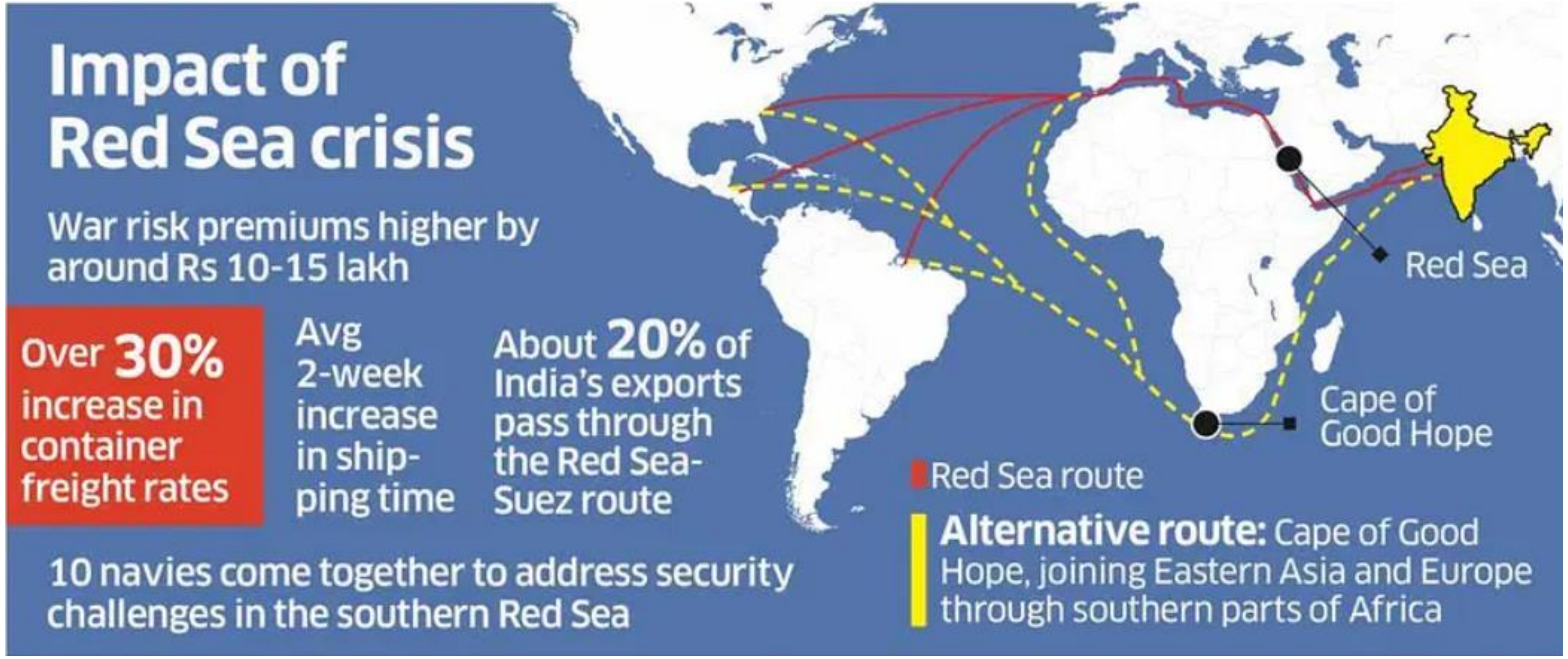
Examples: Material Management and Geopolitics

Why China is working on One Belt One Road program?

[Belt and Road Initiative - Wikipedia]

Examples: Material Management and Geopolitics

Red Sea Crisis?

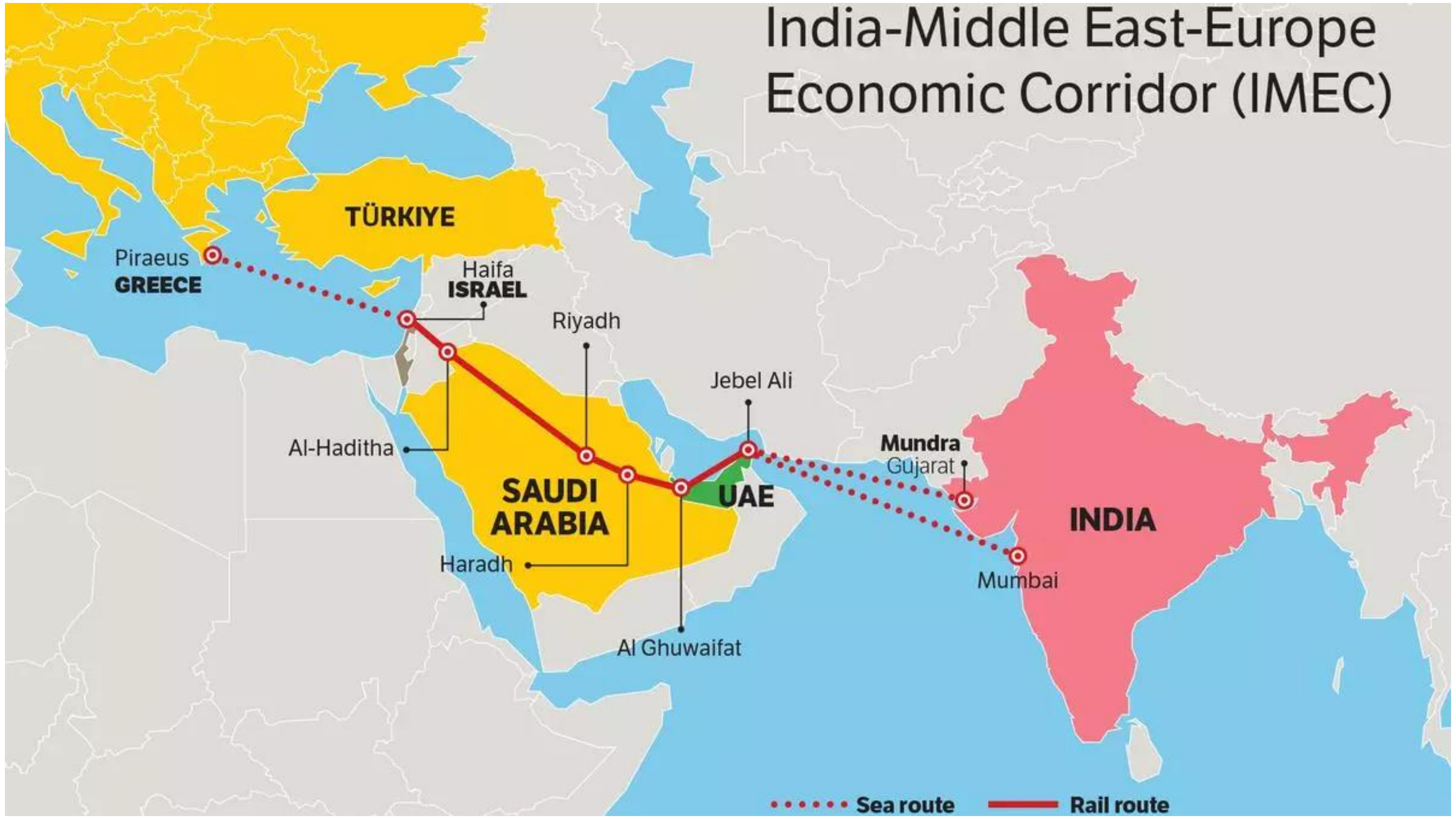


[Red Sea crisis - Wikipedia]

Examples: Material Management and Geopolitics

How to overcome Red Sea Crisis?

India-Middle East-Europe Economic Corridor

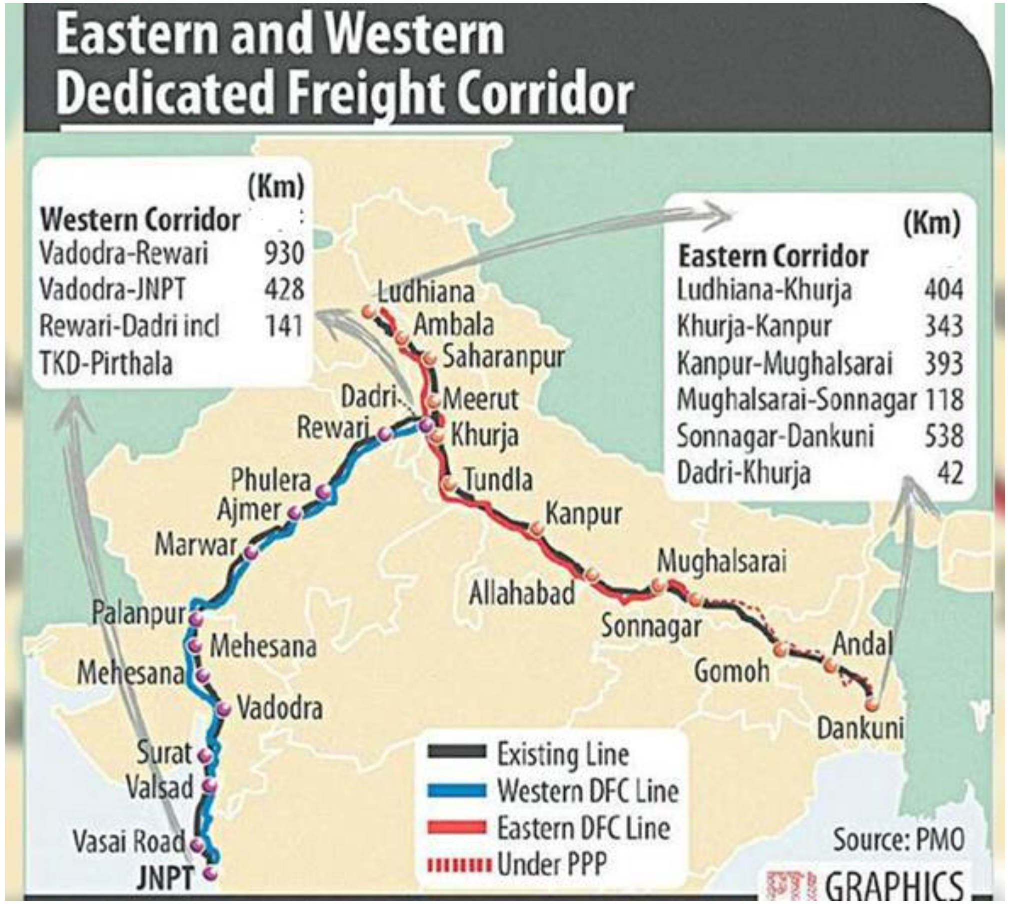


India-Middle East-Europe Corridor (drishtiias.com)

Factors Providing Economy in Material Management

1. Volume of purchases
2. Nearness to sources of materials
3. Design and engineering of the product
4. Inspection
5. Make or buy decision
6. Disposal of scrap

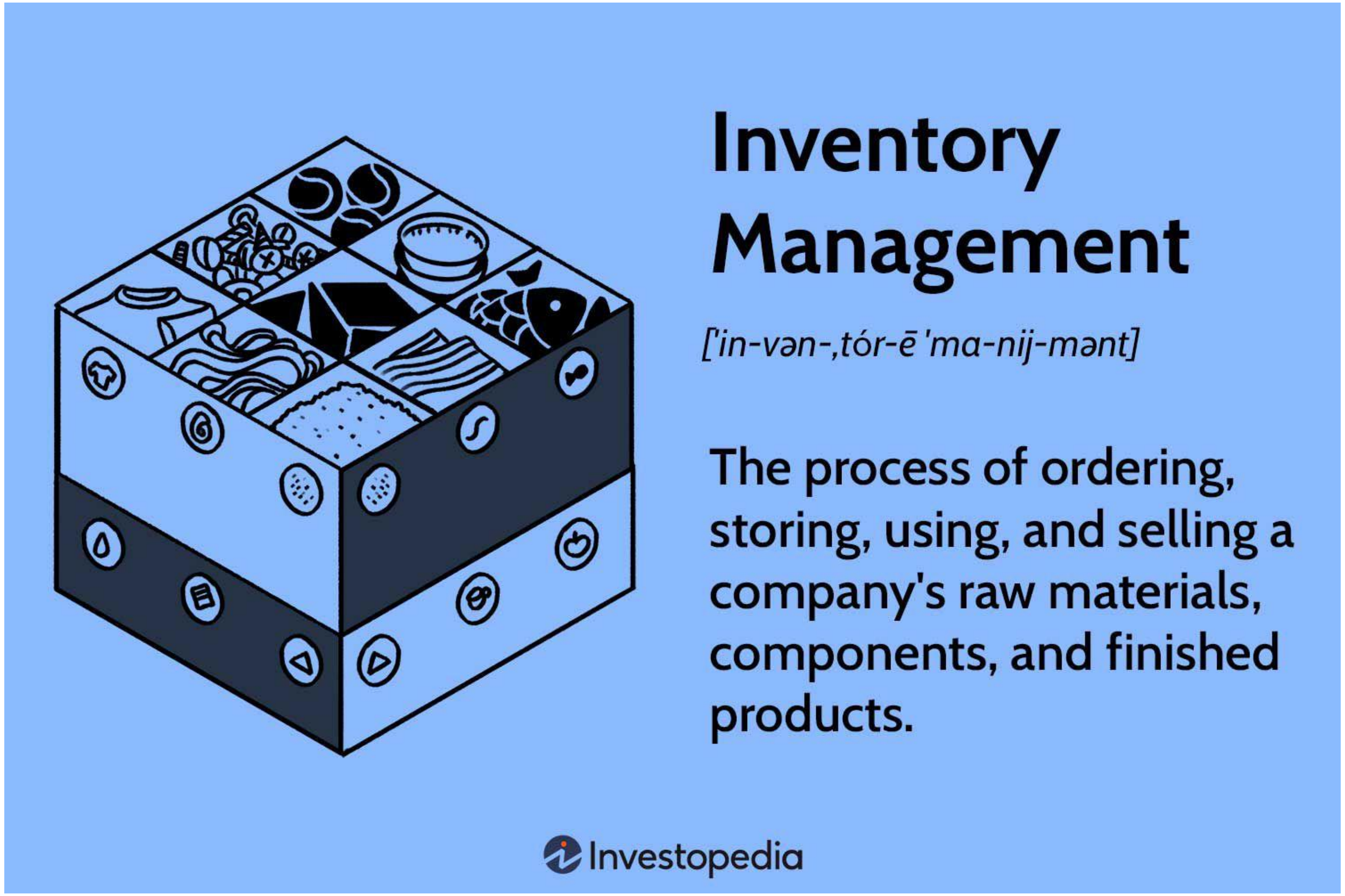
Example: Dedicated Freight Corridor



Dedicated Freight Corridors are planned to reduce cost as well as time for material transport with in mainland

[Dedicated freight corridors: Transforming India's logistics backbone (itln.in)](#)

Inventory Management




What is Inventory Management?

Inventory Control

- Inventory is the **list of movable goods which directly or indirectly used in the production of goods for sale**
- Inventory is **money kept in the store room** in the form of **raw material, in-process material and finished goods**
- “**Inventory control refers to the process whereby the investment in materials and parts carried in stock is regulated within the predetermined limits set in accordance with the inventory policy established by the management**”
- It is the **scientific method of finding how much stock should be maintained in order to meet the production demands at the minimum cost**

Inventory Control

Classification

- **Direct inventories**
  - Raw materials
  - In process inventories (work in progress)
  - Purchased parts
  - Finished goods
- **Indirect inventories** - will not be part of product
  - Tools:- machine tools and hand tools
  - Supplies:- accessories of machines, cleaning materials, lubricants, office stationary.

Need for Inventories (Functions or Advantages)

- To ensure against delays in deliveries due to variation in lead time
- To allow for possible increase in output due to variation in demand
- Maintain smooth and efficient production flow
- To keep better customer relations
- To take advantage of quantity discounts
- To utilize advantage of price fluctuations
- To ensure against scarcity of materials in the market
- To have better utilization of men and machinery

Determining Inventory Level

- Objective is to **minimize inventory cost** and **maximize profit**
- Variables which affect level of inventory are
    - **Order quantity**
    - **Lead time**
    - **Safety stock**
    - **Reorder point**

Determining inventory level

- **Order quantity**
  - The quantity of materials ordered in a single order. It should be decided after considering cost (EOQ)
- **Lead time**
  - Time Interval between **initiating the order and the material is actually received**
  - It consists of
    - **time taken to deliver purchase order to seller**
    - time for the seller to get or prepare the inventory for dispatch
    - time taken for transportation to and reception of inventory by the customer

Determining inventory level

- **Safety stock**
  - The use rate and lead time are varying because supplier may fail to keep delivery promise the forecast of use rate may be inaccurate
  - So **extra inventory is needed in order to avoid shortage**. This is called **reserve stock/safety stock/buffer stock**
  - It should be optimum depending upon
    - Lead time demand
    - Carrying charges
    - Importance of items

Determining inventory level

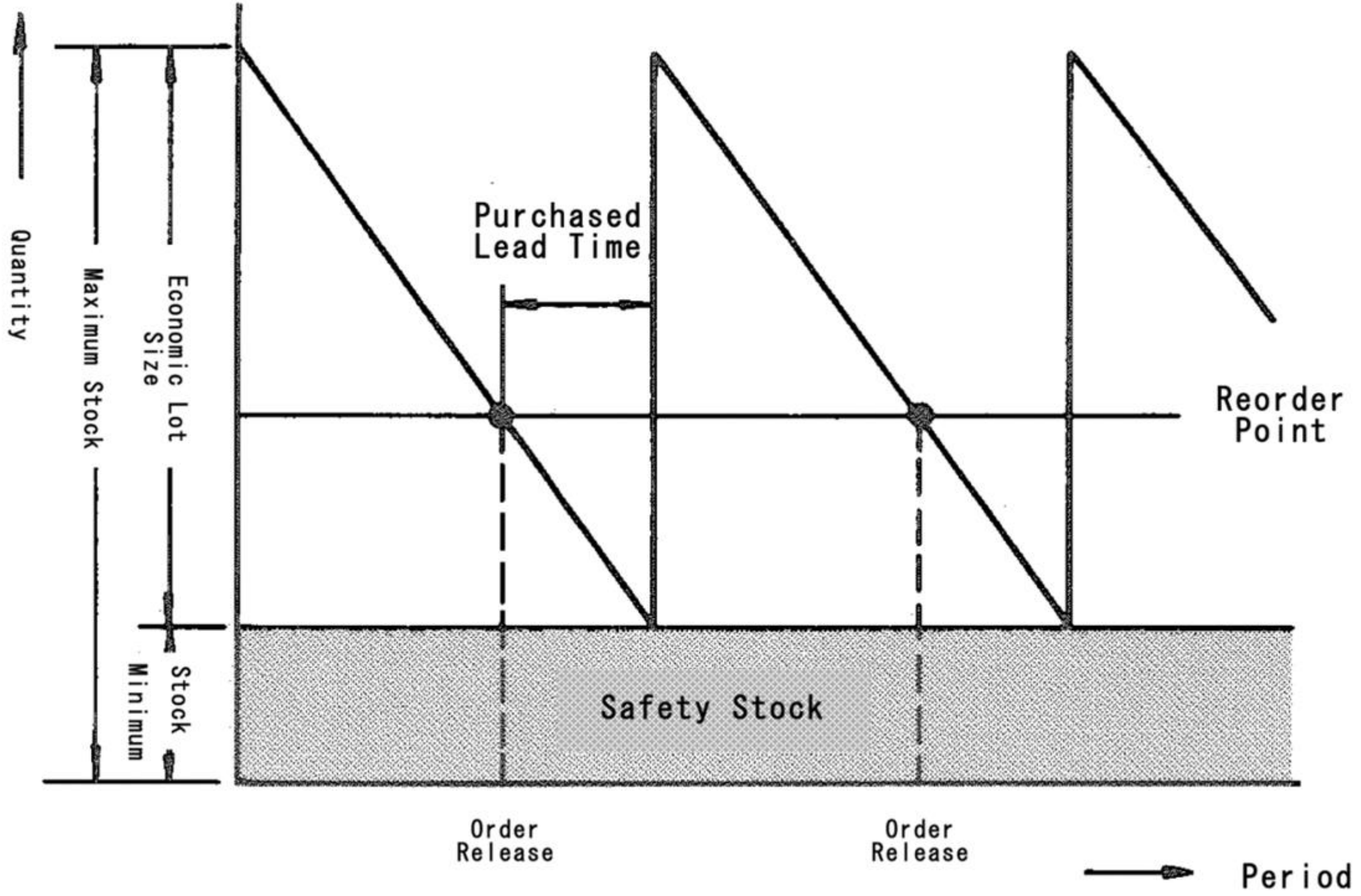
- **Reorder point**
  - This indicates the time (**level of inventory) at which the purchase order should be initiated and if not done so, the production may stop due to shortage**

Eg: Reorder point = safety stock + (lead time *consumption rate)

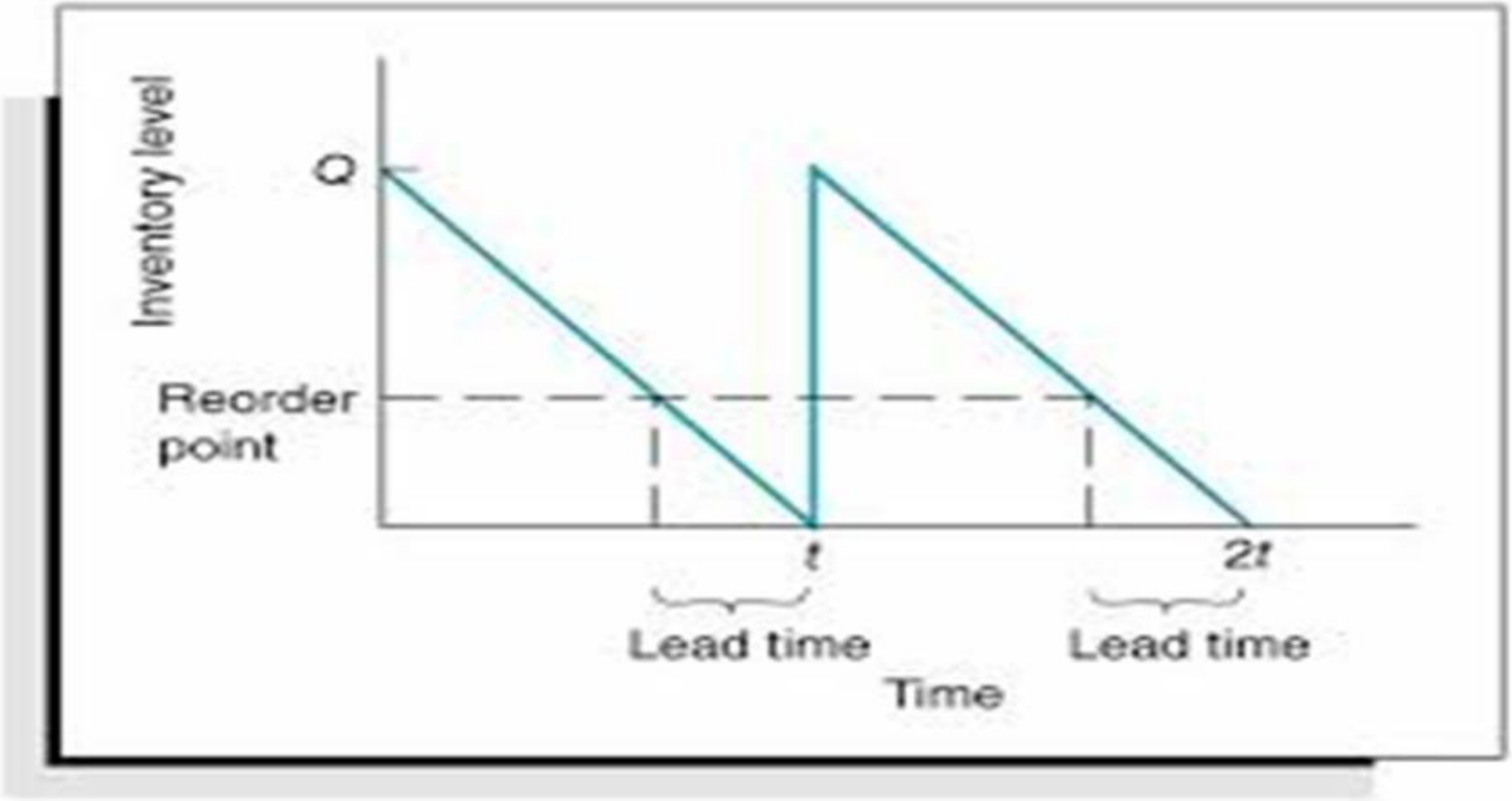
= 100 units + (1 month* 200 units/month)

= 300 units

Inventory Vs Time



Inventory Vs Time (Without safety stock)



Inventory Models

- Inventory Models seek to find the best balance between finance, production & marketing.
- **When to order and how much to order**
- Two models mainly,
    1. Static Inventory Models
    2. Dynamic Inventory Models

Inventory Models

Static Inventory Models

- Static applies when one decision is allowed, about how much to buy for a constrained marketing window such as seasonal goods, perishable goods like vegetables, bread etc.
- In simple, **only one order can be placed to meet the demand**
- **Repeat orders are either impossible or too expensive**

Inventory Models

Dynamic Inventory Models

- Dynamic situations require **continuous decisions** about **how much to order at different points in time, ie. repeat orders can be placed**

  a) Deterministic models

  - demand and lead time of an item is constant, ie. known exactly
  - Purchase model
  - Production inventory model

  b) Probabilistic models

  - Variation in demand and lead time

Inventory Cost

Total inventory cost = Ordering cost + Carrying cost (Holding cost)

Cost of ordering = C_o (Rs)

Order quantity =q

Unit cost of ordering = C_o / q

If 's' is the annual demand, then

Annual Ordering cost= C_o / q * s

Annual Carrying cost =Cu * i * q/2

Cu = Unit purchase cost

i = Interest rate

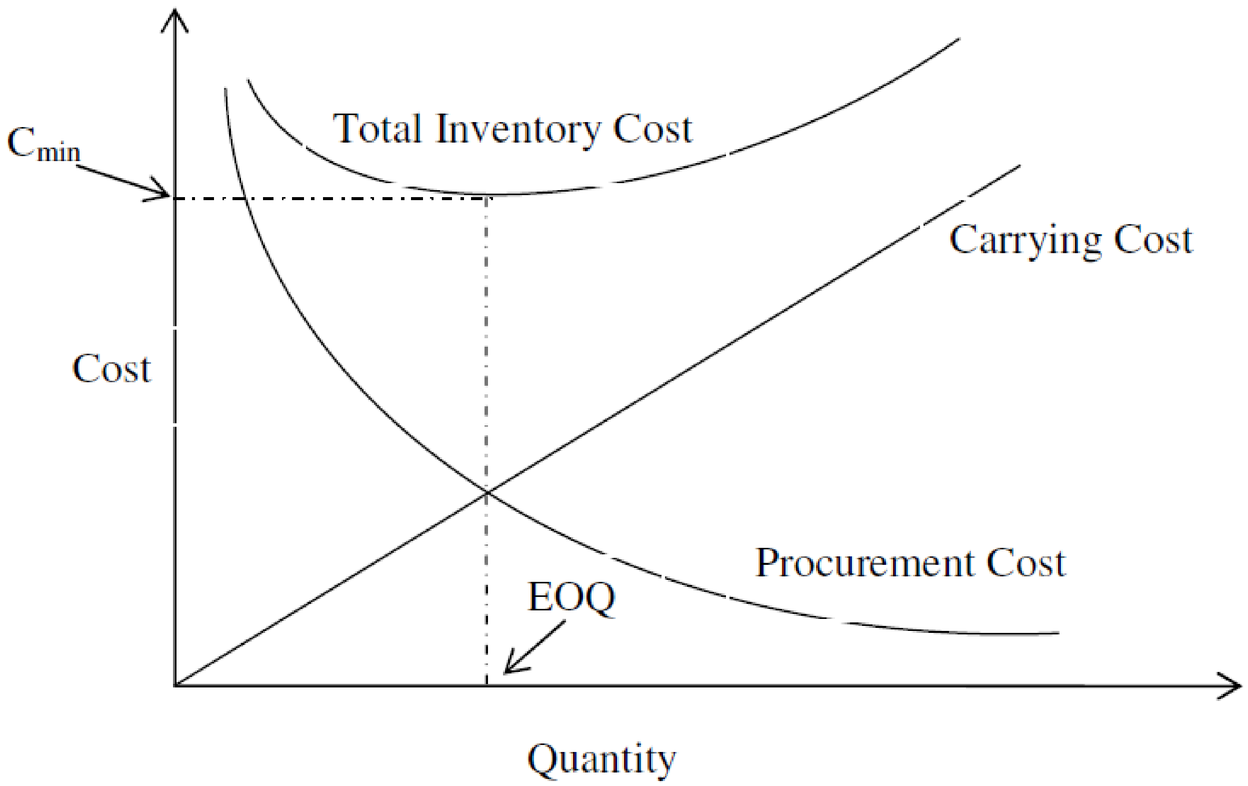
q/2 = avg. inventory level

Economic Order Quantity (EOQ)

- The economic **order quantity is the size of an inventory order which minimizes the inventory cost. The inventory cost is the sum of procurement cost and carrying cost**
- To determine EOQ two extreme views are encountered:
  - **Order for very large lots** (Produce in very large lots) **to minimize the procurement cost (to minimize set up cost)**
  - **Order for every small lots** (produce in very small lots) **to minimize the storage cost or carrying cost.**

Economic Order Quantity (EOQ)

- Purchasing Model with No Shortage
- Assumptions
  - **Fixed demand rate (fixed production rate)**
  - **Instantaneous replacement (Lead time = 0)**
  - **No shortage is permitted.**
- Procurement cost = Carrying cost



Economic Order Quantity (EOQ)

Let, C_3 = Procurement cost / order (Setup cost / order)

r = Demand rate in units / year

C_1 = Carrying cost / unit / year *(C_1 is generally expressed as a fraction of cost of unit)*

q = Quantity of units ordered at time t

(Note: One year is considered as the unit time)

$$\text{No. of orders placed in a year} = \frac{r}{q}$$

$$\text{Therefore, Procurement cost / year} = \left(\frac{r}{q}\right)C_3$$

$$\text{Total carrying cost / year} = \left(\frac{q}{2}\right)C_1$$

Economic Order Quantity (EOQ)

Therefore, Inventory cost/ year, C = procurement cost / year + carrying cost / year

$$C = \left(\frac{r}{q}\right)C_3 + \left(\frac{q}{2}\right)C_1$$

To minimize 'C', $\frac{dC}{dq} = 0$

$$\Rightarrow -\frac{rC_3}{q^2} + \frac{C_1}{2} = 0$$

$$\Rightarrow q = \sqrt{\frac{2C_3 r}{C_1}} = EOQ \text{ (Economical Order Quantity)}$$

Substituting the value of EOQ in the cost equation, $C = \left(\frac{r}{q}\right)C_3 + \left(\frac{q}{2}\right)C_1$, the minimum inventory cost can be expressed as $C_{min} = \sqrt{2C_1C_3r}$

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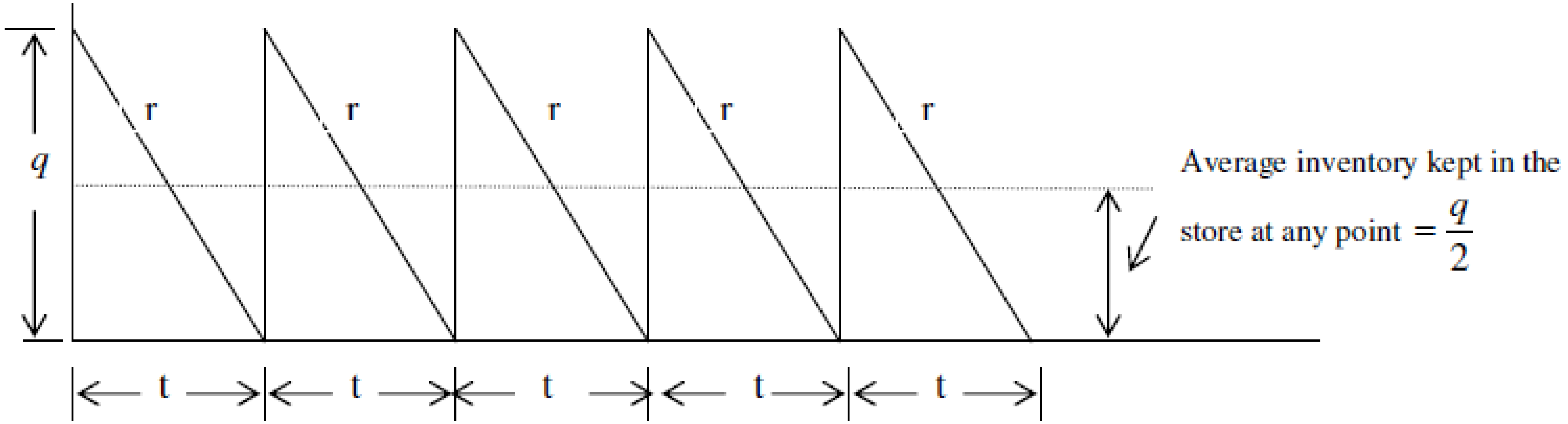
Economic Order Quantity (EOQ)



$$\text{Economic Order Quantity, } EOQ = \sqrt{\frac{2C_3 r}{C_1}}$$

$$\text{Minimum Cost of Inventory, } C_{\min} = \sqrt{2C_1C_3r}$$

Economic Order Quantity (EOQ)

Problem

A manufacturer has to supply his customers with 600 units of his products per year. Shortages are not allowed and the storage cost amounts to ₹ 0.60 per unit per year. The setup cost per run ₹ 80. Find the optimum run size, its number and the minimum yearly cost of inventory.

Economic Order Quantity (EOQ)

Problem

A manufacturer has to supply his customers with 600 units of his products per year. Shortages are not allowed and the storage cost amounts to ₹ 0.60 per unit per year. The setup cost per run ₹ 80. Find the optimum run size, its number and the minimum yearly cost of inventory.

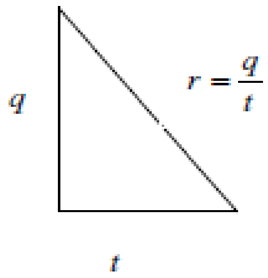
Answer

$C_1 = ₹\ 0.60 / \text{year}$

$C_3 = ₹\ 80$

$r = 600 \text{ units / year}$

Optimum run size, $EOQ = \sqrt{\dfrac{2C_3 r}{C_1}} = 400 \text{ units}$



$r = \dfrac{q}{t}$ $\qquad r = \dfrac{q}{t} \Rightarrow t = \dfrac{q}{r}$

$$t = \frac{q}{r} = \frac{400}{600} = \frac{2}{3}\ years = \frac{2}{3} \times 12 = 8 \cdot months$$

Minimum Cost of Inventory, $C_{\min} = \sqrt{2C_1C_3r} = ₹\ 240 / \ year$

That is, the manufacturer has to produce **400 units** of his products once in **8 months** and the minimum inventory cost works out to be **₹ 240 / year.**

Material Requirement Planning (MRP)

- Material Requirement Planning (MRP) refers to the basic calculations used to determine component requirements from end item requirements.
- It also refers to a broader information system that uses the dependence relationship to plan and control manufacturing operations.
- MRP is a technique for determining the quantity and timing for the acquisition of dependent demand items needed to satisfy master production schedule requirements.

Material Requirements Planning (MRP)

- Computer-based information system for ordering and scheduling of inventories, i.e. what is needed, how much is needed, and when is it needed

- Functions of MRP system
  - Control of inventory levels
  - Assignments of priorities for components
  - Determination of capacity requirements at a detailed level

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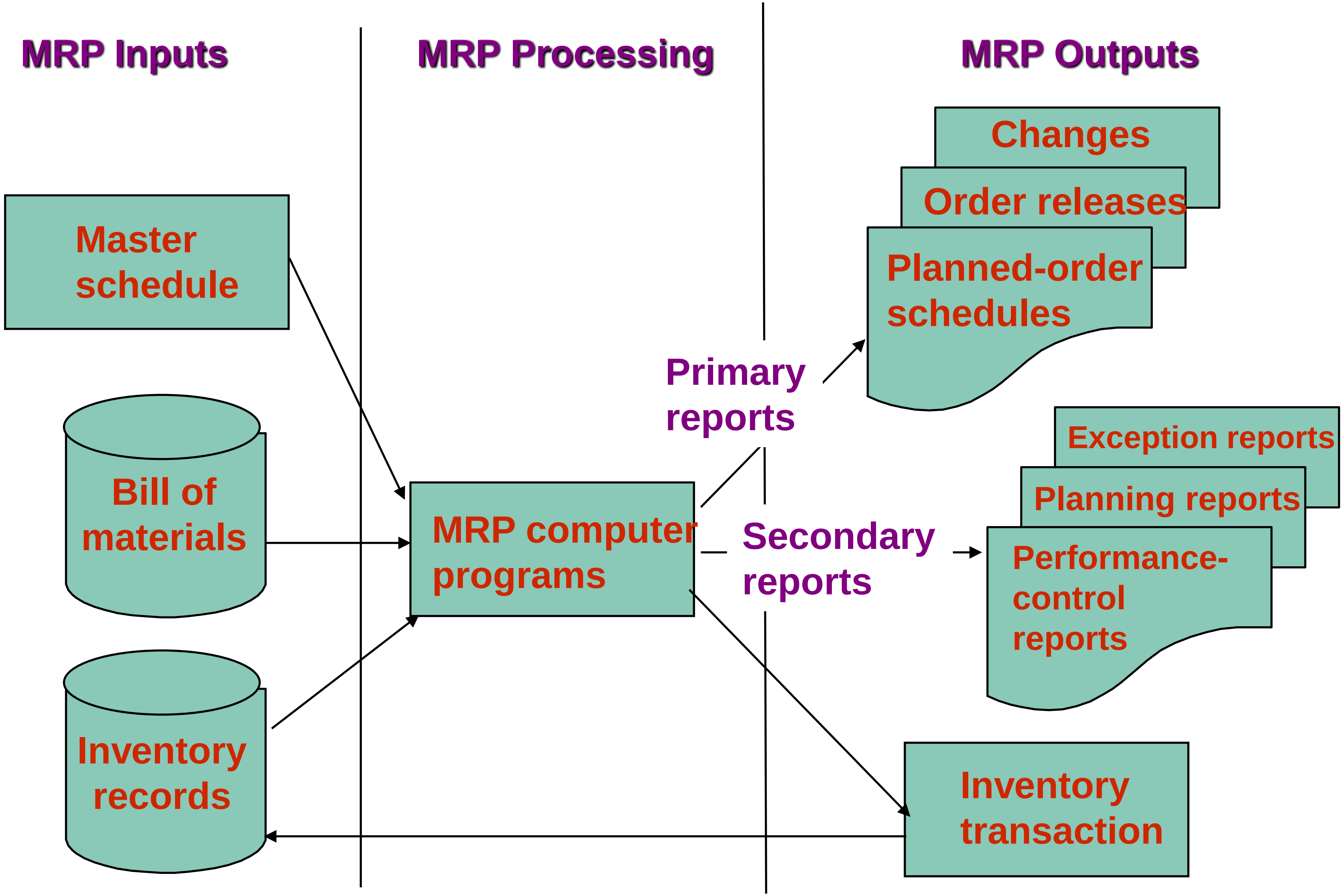
Objectives of MRP

- Inventory reduction
- Reduction in the manufacturing and delivery lead times
- Realistic delivery commitments
- Increased efficiency


MRP Inputs
MRP Processing
MRP Outputs
Changes
Order releases
Planned-order schedules
Master schedule
Primary reports
Bill of materials
Exception reports
Planning reports
MRP computer programs
Secondary reports
Performance-control reports
Inventory records
Inventory transaction

MRP Inputs

- **Master Production Schedule (MPS)** – States which end items are to be produced, when they are needed, and in what quantities. It controls the timing and quantity of production for products or product families
- **Bill of Materials (BOM)** – a listing of all of the raw materials, parts, and sub-assemblies needed to produce one unit of a product
- **Inventory Records** – includes information on the status of an item during the planning horizon, eg. quantity, supplier, order lead time, lot size

MRP Outputs

Primary Reports

- ***Planned Orders Schedule*** - indicating the amount and timing of future orders
- ***Order Releases*** - Authorization for the execution of planned orders
- ***Change***s - revisions of due dates or order quantities, or cancellation of orders

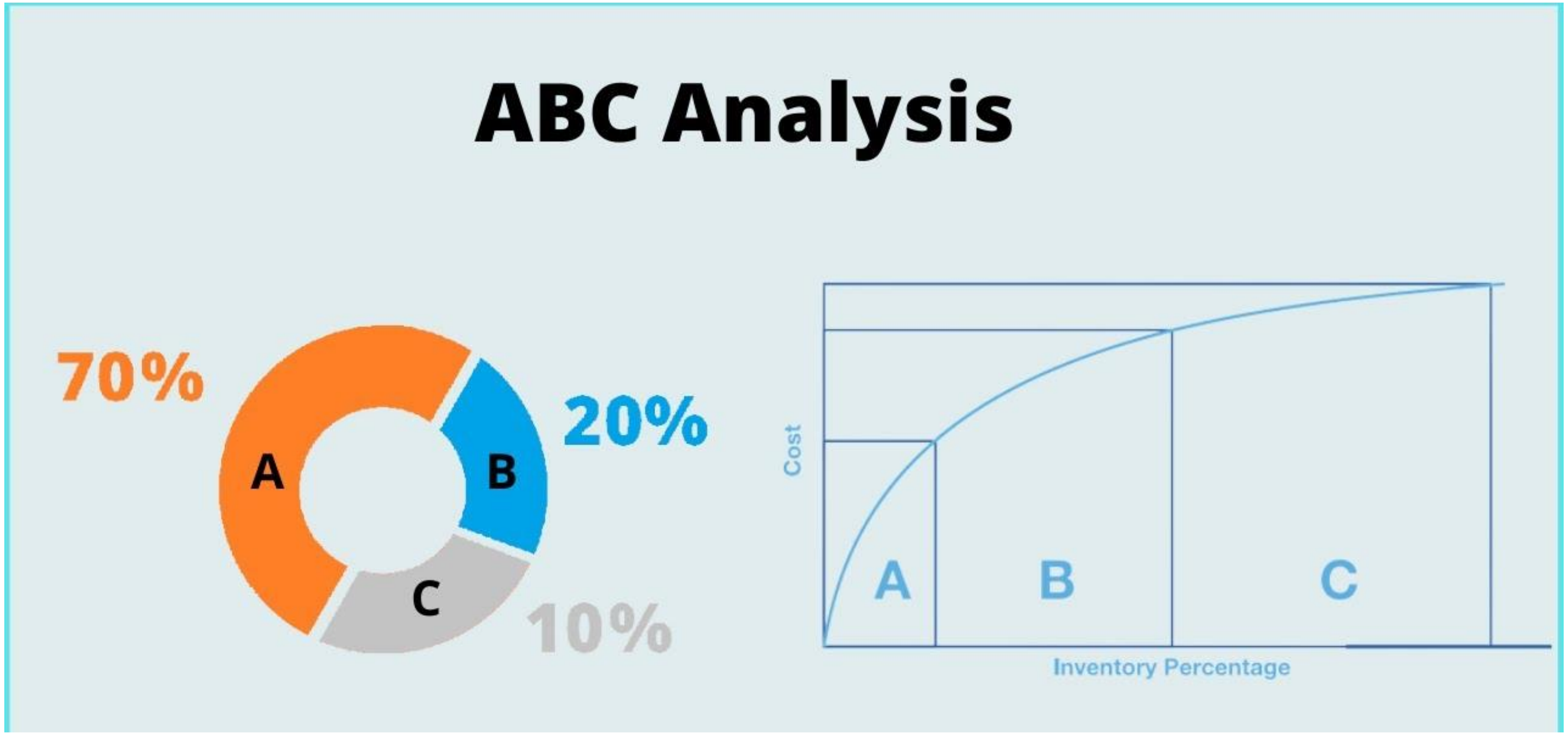
Secondary Reports

- ***Performance-control reports*** - Evaluation of system operation, including deviations from plans and cost information
- ***Planning reports*** - Data useful for assessing future material requirements
- ***Exception Reports*** - Data on major discrepancies encountered

Benefits of MRP

- Improved customer service
- Quicker response to changes in demand
- Better machine utilization
- Greater productivity
- Reduction in lead time
- Reduction in work- in- process
- Reduction in past due orders
- Elimination of annual inventory
- Reduction in safety stock
- Increase in productivity
- Capacity constraints are better understood

ABC Analysis



ABC Analysis

- ABC analysis, also known as Selective Inventory Control is a business term used to define an inventory categorization technique often used in materials management.
- It can be observed that in a typical industrial situation nearly 10% of items have 70% of the annual inventory consumption, 20% of the items have 20% of annual inventory consumption, and 70% of the items have only 10% of the annual inventory consumption.

ABC Analysis

- Since 70% of the annual consumption of inventory is covered by only 10% of the items in the inventory, these items deserve highest attention and are classified as ‘A’ items.
- Similarly 20% of the items covering 20% of the inventory investment are B class items and require moderate attention.
- Balance 70% of the inventory items are termed as C class items and require least attention as .they consume only 10% of the total inventory cost.

Steps in ABC Analysis

1) Determine the annual usage in units for each item for the past one-year.
2) Multiply the annual usage quantity with the average unit price of each item to calculate the annual usage for each item in monetary terms.
3) Item with highest money usage annually is ranked first. Then the next lower annual usage item is listed till the lowest item is listed in the last.
4) Arrange the items in the inventory by cumulative annual usage (Rupees) and by cumulative percentage. Categorize the items in A, B, and C categories.

Advantages of ABC Analysis

- A Close and strict control is facilitated on the most important items which constitute a major portion of overall inventory valuation or overall material consumption & due to this, costs associated with inventories maybe reduced.
- The investment in inventory can be regulated in proper manner & optimum utilization of available funds can be assured.
- A strict control on inventory items in this manner helps in maintaining a high inventory turnover rates.

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Project Management

Project Management

- Project management is the application of processes, methods, skills, knowledge and experience to achieve specific project objectives according to the project acceptance criteria within agreed parameters. Project management has final deliverables that are constrained to a finite timescale and budget
- A key factor that distinguishes project management from just 'management' is that it has this final deliverable and a finite time span, unlike management which is an ongoing process. Because of this a project professional needs a wide range of skills; often technical skills, and certainly people management skills and good business awareness.

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Project Manager

- In the broadest sense, **project managers (PMs)** are responsible for planning, organizing, and directing the completion of specific projects for an organization while ensuring these projects are on time, on budget, and within scope

Functions - Project Management

- Managing the risks, issues and changes on the project;
- Monitoring progress against plan;
- Managing the project budget;
- Maintaining communications with stakeholders and the project organisation;
- Closing the project in a controlled fashion when appropriate.

Characteristics - Project Management

Characteristics - Project Management

- Scope: defines what will be covered in a project.
- Resource: what can be used to meet the scope.
- Time: what tasks are to be undertaken and when.
- Quality: the spread or deviation allowed from a desired standard.
- Risk: defines in advance what may happen to drive the plan off course, and what will be done to recover the situation.

Feasibility Study

A **feasibility study** is an **analysis** that takes all of a **project's** relevant factors into account—including economic, technical, legal, and scheduling considerations—to ascertain the likelihood of completing the **project** successfully.

Project Management

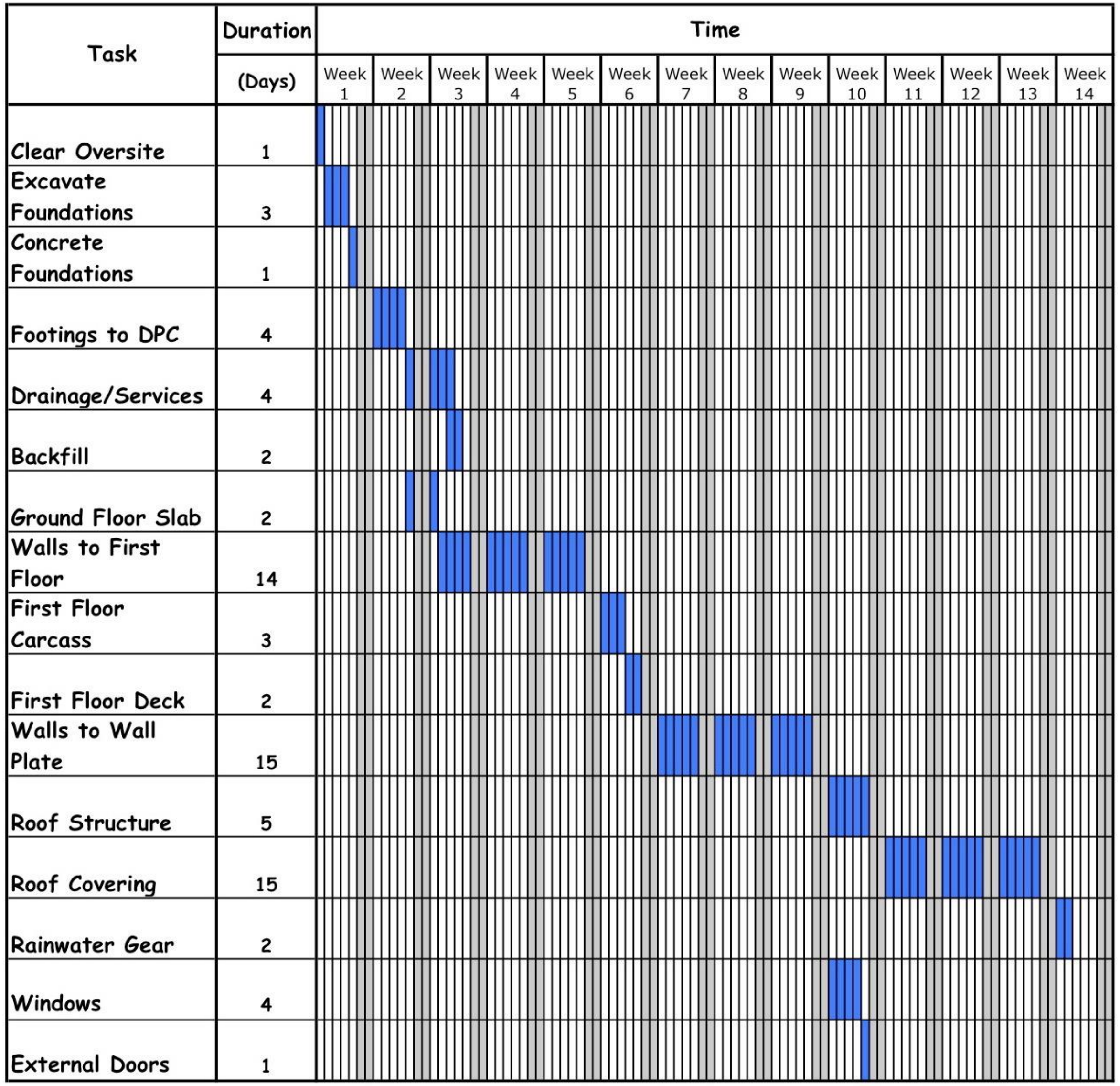
Project Management

- Process of **planning, organizing and controlling resources** to achieve specific project objectives within a **defined scope, time frame and budget**
- Involves overseeing a team **to complete task, manage tasks, and ensure the project meet its goal**

Example: Building a house

Level				
Level 1	House to Watertight Shell			
Level 2	Groundworks	Superstructure	Roof	Windows and Doors
Level 3	Remove Oversite Excavate Foundations Concrete Foundations Footings to DPC Drainage/Services Backfill	Ground Floor Walls to First Floor First Floor Carcass First Floor Deck Walls to Wall Plate	Roof Structure Roof Covering Rainwater Gear	Windows External Doors

Example: Building a house- Gantt Chart



Project Management in Metro Rail Construction

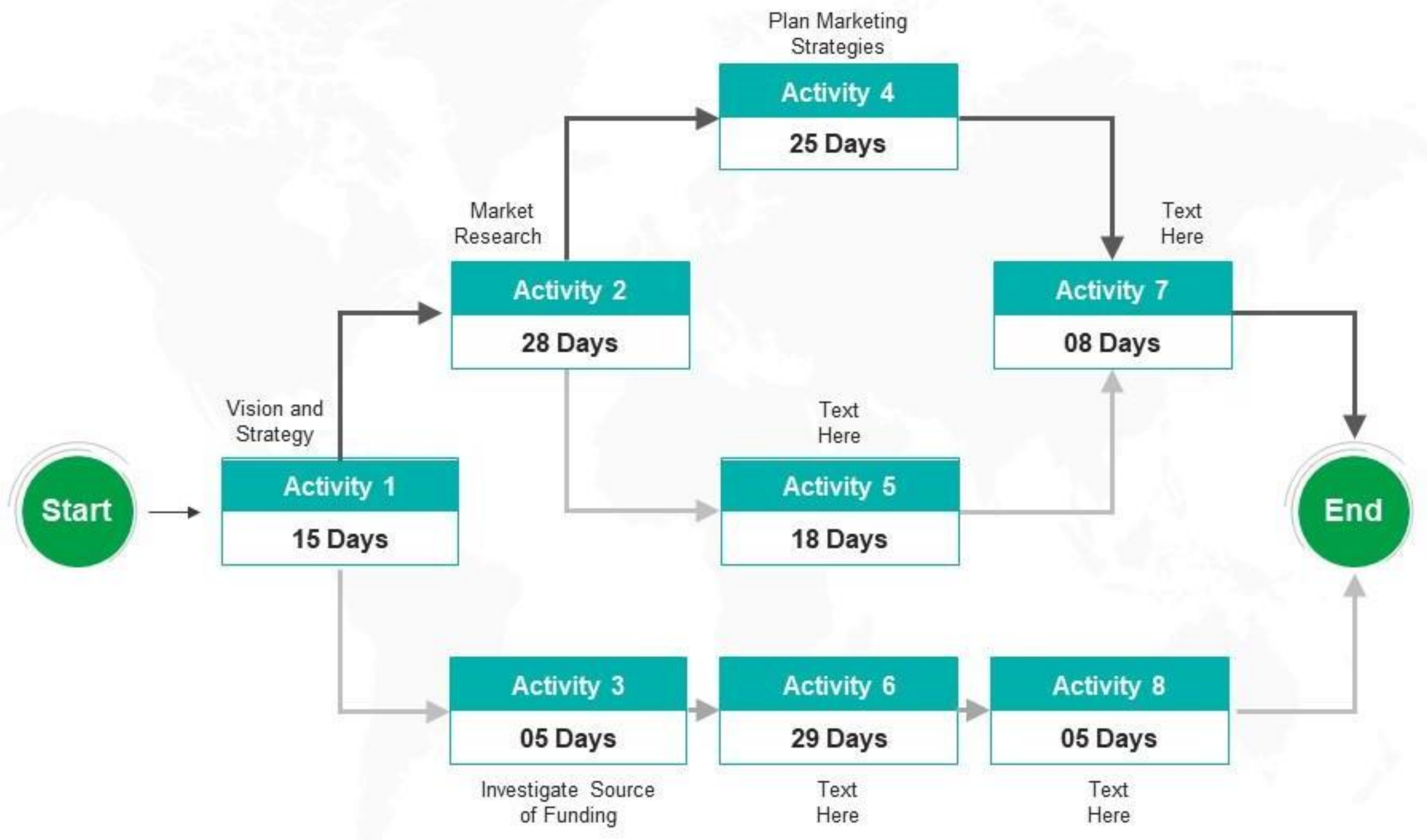
- **Pre-construction Phase**
  - Feasibility study
  - Route selection
  - Environmental impact assessment
  - Land acquisition
  - Design and Engineering, Detailed plan
- **Construction Phase**
  - Site preparation
  - Foundation work
  - Civil work
  - Track laying
  - Electrical and Signal system
  - Station construction
  - Depot construction

[Metro Phase Ii To Be Ready By Nov 2026: Union Min | Kochi News - Times of India (indiatimes.com)]

Project Network Analysis

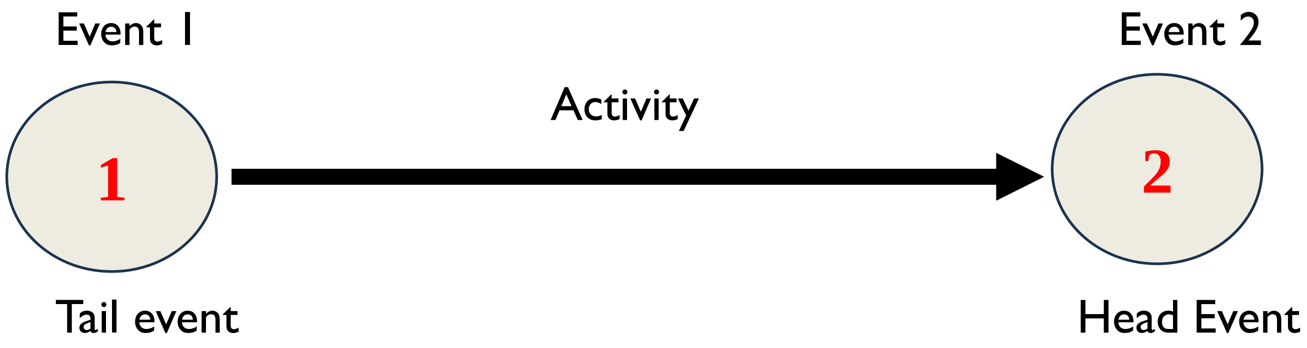
- Technique used in project management **to visualize and analyze the sequence of tasks, their dependence, critical path within the project**
- Helps in **identifying potential bottlenecks, assessing project duration and evaluating the impact of changes on project time line**
- Two methods
  - **Critical path method (CPM)**
    - Single estimate of activity time
    - Deterministic activity time
  - **Project Evaluation and Review Technique (PERT)**
    - Multiple time estimate
    - Probabilistic activity time

Project Network Diagram



Network diagram representation

- **Activity**:- Individual operations which utilizes resources and has an end and beginning, some types are
  - Predecessor activity
  - Successor activity
  - Concurrent activity
  - Dummy activity
- **Events**:- Point in time signifying completion of some activities and beginning of new one. Usually indicated with circle and number



Critical Path Method

- **Critical path**:- Strongest path in a chain , if the link in critical path breaks/weakens the entire chain will break/entire project delays
- **Relevance of CPM** in Project Management
  - **Identifies Critical Tasks**: CPM highlights the tasks that are essential for the project's completion and cannot be delayed without affecting the overall timeline.
  - **Focuses on Key Activities**: By understanding the critical path, project managers can prioritize resources and attention on the most important tasks, ensuring they are completed efficiently.
  - **Helps in Resource Allocation**: CPM aids in allocating resources effectively by showing which tasks require more attention and resources to ensure they are completed on time.
  - **Facilitates Risk Management**: By identifying the critical path, potential risks and bottlenecks can be more easily identified and addressed.
  - **Enhances Decision Making**: CPM provides a clear picture of the project's timeline, allowing for informed decisions regarding resource allocation, scheduling, and risk mitigation.

CPM:- Procedure

1. Draw network diagram and indicate events, activities properly
2. Forward pass computations;- computation begins from start note and move towards end node. Identify earliest event time for all events.

$(E_{head})_{i,j} = (E_{tail})_{i,j} + D_{i,j}$, where E indicates Earliest Event Start time, and D indicates activity duration. For junction events, take maximum value of E

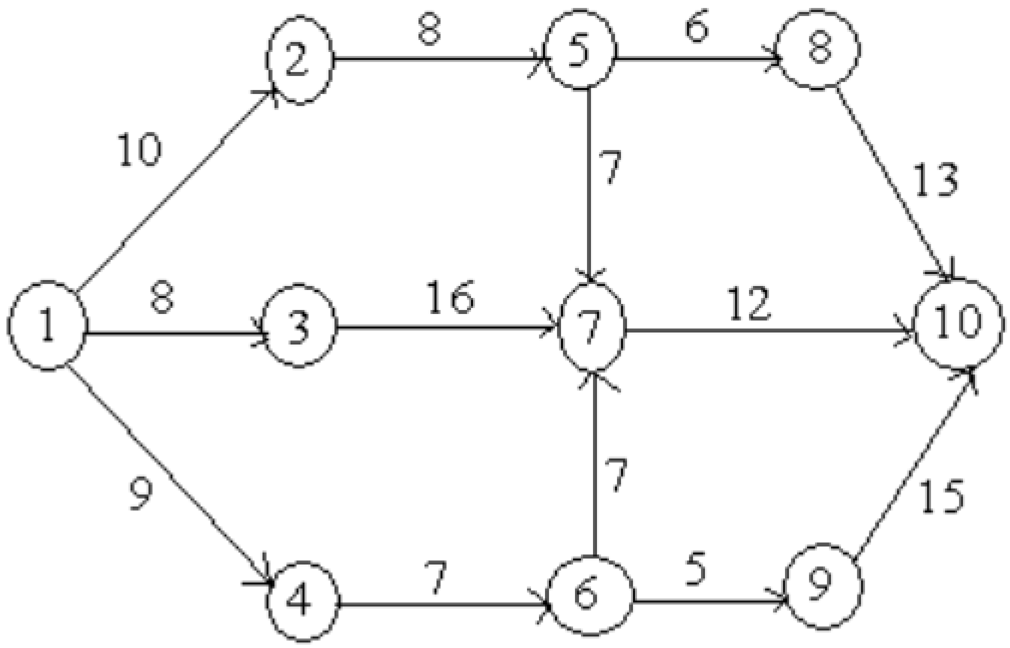
3. Backward pass computation:- For ending event take E = L, Calculate latest starting time from Ending activity to Starting activity.

$(L_{tail})_{i,j} = (L_{head})_{i,j} - D_{i,j}$, For junctions, take minimum value of L

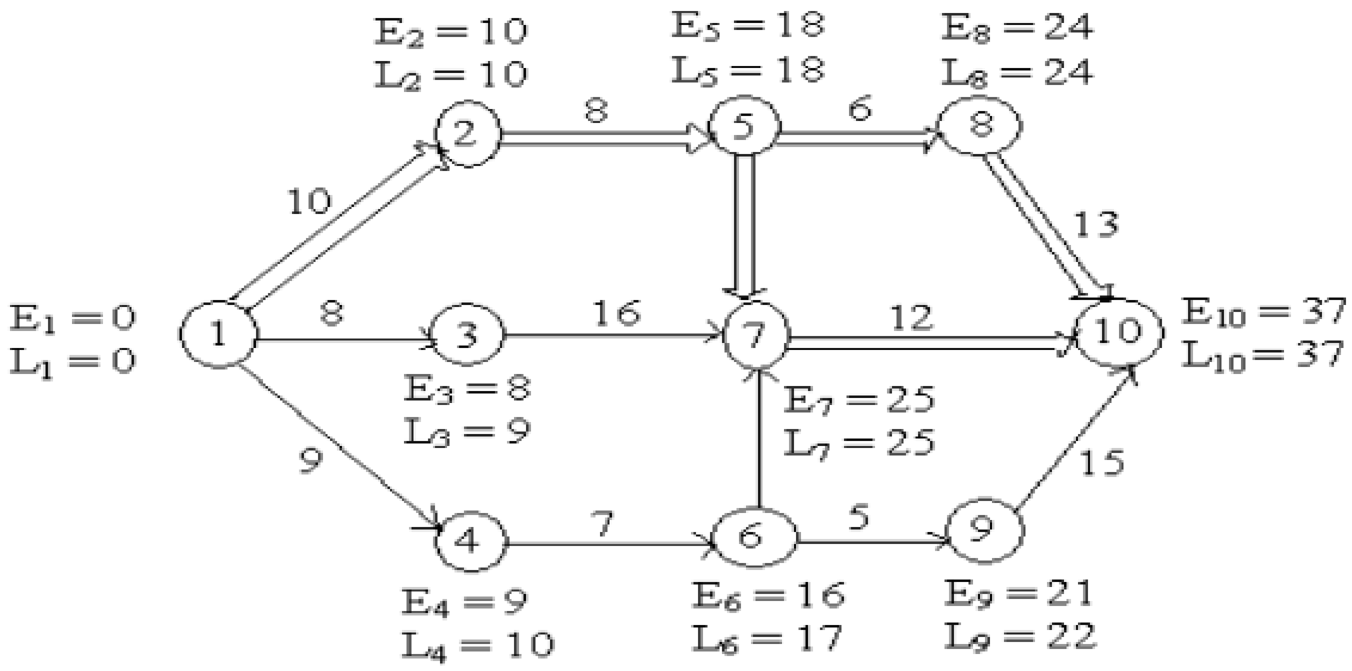
4. Determine Float = L-E for each event
5. Idenfity CPM by connecting Float = 0 events

Example- I

Determine the early start and late start in respect of all node points and identify critical path for the following network.



Example-1 Solution



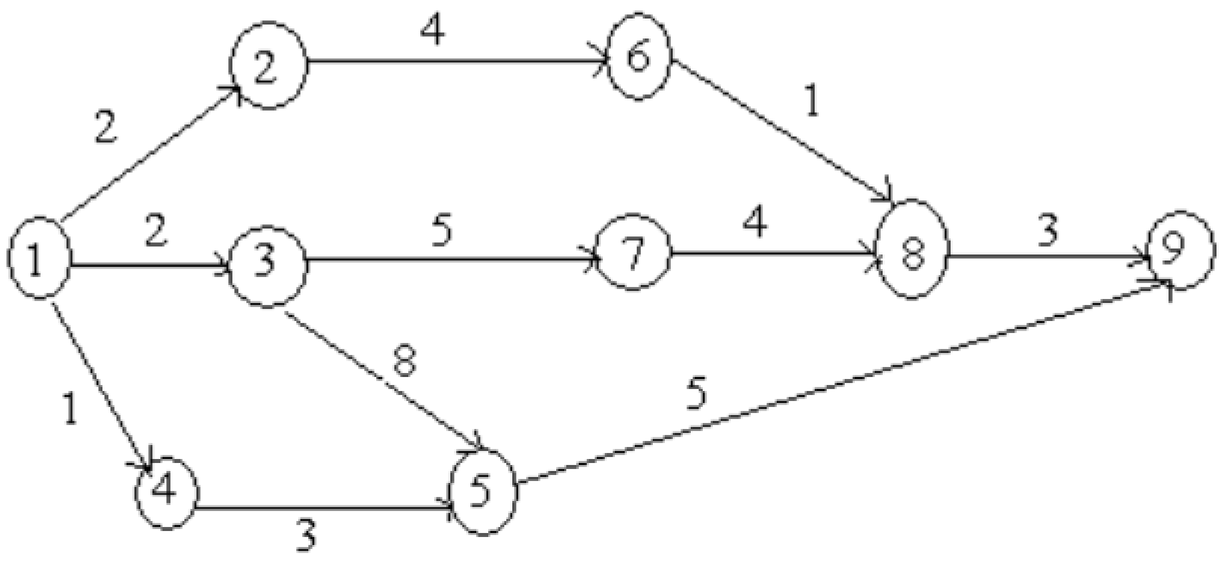
Example-1 Solution…

Activity(i, j)	Normal Time (D_{ij})	Earliest Time		Latest Time		Float Time $(L_i - D_{ij}) - E_i$
		Start (E_i)	Finish ($E_i + D_{ij}$)	Start ($L_i - D_{ij}$)	Finish (L_i)	
(1, 2)	10	0	10	0	10	0
(1, 3)	8	0	8	1	9	1
(1, 4)	9	0	9	1	10	1
(2, 5)	8	10	18	10	18	0
(4, 6)	7	9	16	10	17	1
(3, 7)	16	8	24	9	25	1
(5, 7)	7	18	25	18	25	0
(6, 7)	7	16	23	18	25	2
(5, 8)	6	18	24	18	24	0
(6, 9)	5	16	21	17	22	1
(7, 10)	12	25	37	25	37	0
(8, 10)	13	24	37	24	37	0
(9, 10)	15	21	36	22	37	1

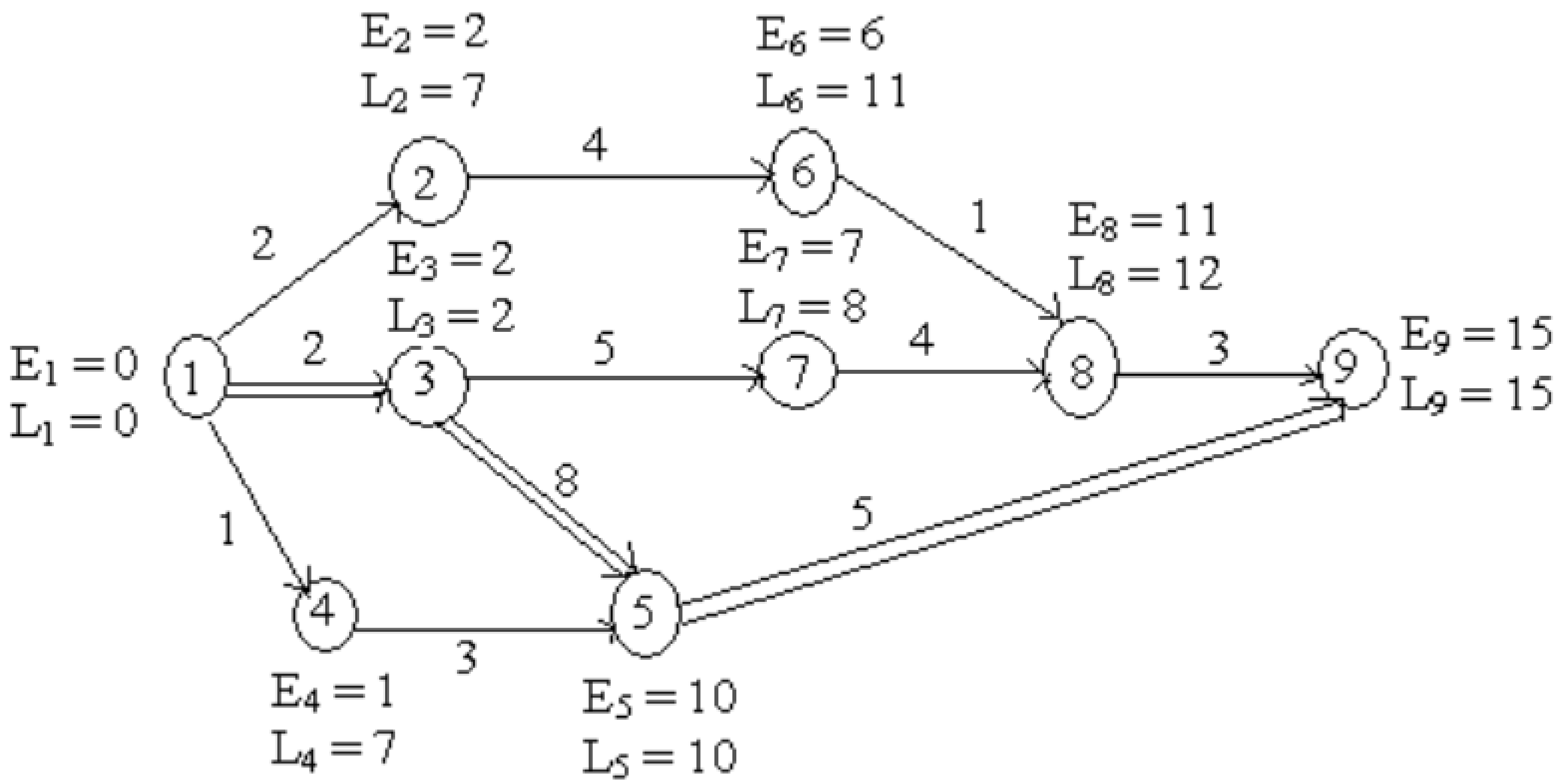
Network Analysis Table

Example-2

Determine the early start and late start in respect of all node points and identify critical path for the following network.



Example-2 Solution



Example-2 Solution…

Activity(i, j)	Normal Time (D_{ij})	Earliest Time		Latest Time		Float Time (L_i - D_{ij}) - E_i
		Start (E_i)	Finish ($E_i + D_{ij}$)	Start (L_i - D_{ij})	Finish (L_i)	
(1, 2)	2	0	2	5	7	5
(1, 3)	2	0	2	0	2	0
(1, 4)	1	0	1	6	7	6
(2, 6)	4	2	6	7	11	5
(3, 7)	5	2	7	3	8	1
(3, 5)	8	2	10	2	10	0
(4, 5)	3	1	4	7	10	6
(5, 9)	5	10	15	10	15	0
(6, 8)	1	6	7	11	12	5
(7, 8)	4	7	11	8	12	1
(8, 9)	3	11	14	12	15	1

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CPM vs PERT

PERT analysis consider uncertainties associated with completion of work!

Project Evaluation and Review Technique (PERT)

- Main objective in the analysis through PERT is to find out the completion for a particular event within specified date
- Takes into account the uncertainties
- Three-time values are associated with each activity
  - **Optimistic time (t_0)** – It is the **shortest possible time** in which the activity can be finished. It assumes that every thing goes very well.
  - **Most likely time (t_m)** – It is the estimate of the normal time the activity would take. This assumes normal delays. If a graph is plotted in the time of completion and the frequency of completion in that time period, then most likely time will represent the highest frequency of occurrence.
  - **Pessimistic time (t_p)** – It represents the **longest time the activity could take if everything goes wrong**.

PERT

- **Expected time** – It is the average time an activity will take if it were to be repeated on large number of times and is based on the assumption that the activity time follows Beta distribution, this is given by

$$t_e = (t_0 + 4\ t_m + t_p) / 6$$

- The variance for the activity is given by

$$\sigma^2 = [(t_p - t_o) / 6]^2$$

- Procedure similar to CPM Method

Example- I

Identify critical path for the following network.

Activity	Most optimistic time (a)	Most pessimistic time (b)	Most likely time (m)
(1 – 2)	1	5	1.5
(2 – 3)	1	3	2
(2 – 4)	1	5	3
(3 – 5)	3	5	4
(4 – 5)	2	4	3
(4 – 6)	3	7	5
(5 – 7)	4	6	5
(6 – 7)	6	8	7
(7 – 8)	2	6	4
(7 – 9)	5	8	6
(8 – 10)	1	3	2
(9 – 10)	3	7	5

Example-1:- Solution…

Task	Least time(t_0)	Greatest time (t_p)	Most likely time (t_m)	Expected time $(to + t_p + 4t_m)/6$

Example-1:- Solution…

Activity	(a)	(b)	(m)	(4m)	t_e (a + b + 4m)/6
(1 – 2)	1	5	1.5	6	2
(2 – 3)	1	3	2	8	2
(2 – 4)	1	5	3	12	3
(3 – 5)	3	5	4	16	4
(4 – 5)	2	4	3	12	3
(4 – 6)	3	7	5	20	5
(5 – 7)	4	6	5	20	5
(6 – 7)	6	8	7	28	7
(7 – 8)	2	6	4	16	4
(7 – 9)	5	8	6	24	6.17
(8 – 10)	1	3	2	8	2
(9 – 10)	3	7	5	20	5

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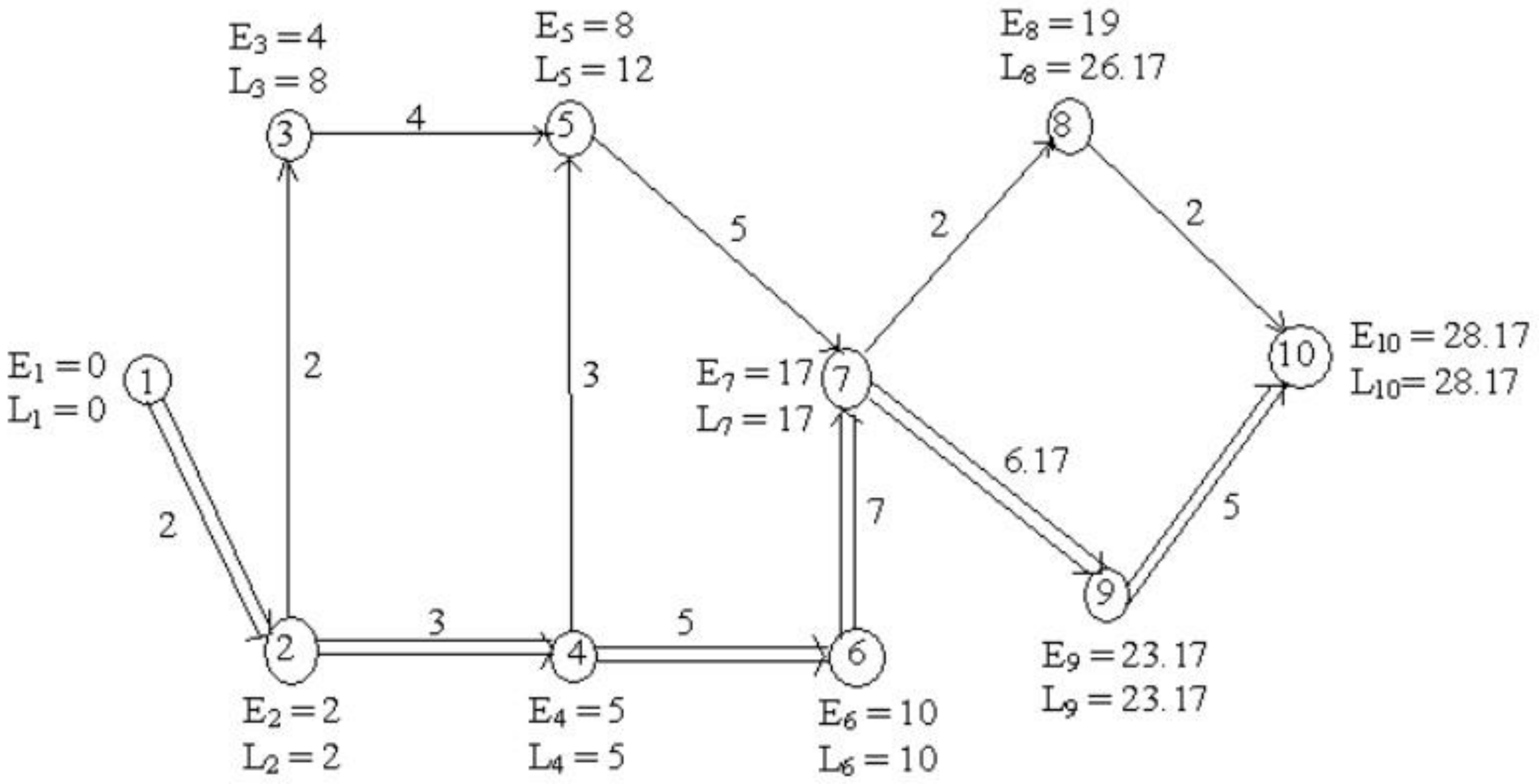
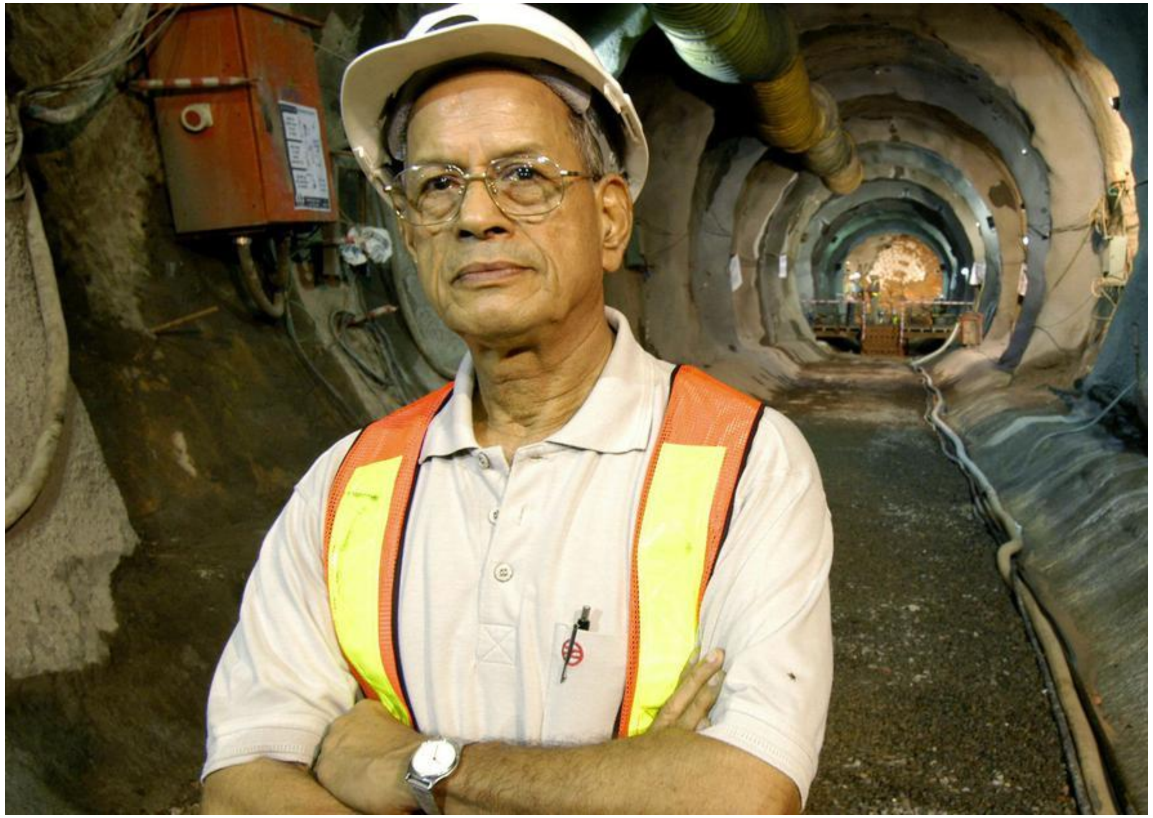
Example-1:- Solution



The critical path = 1 → 2 → 4 → 6 → 7 →9 →10

Example: Project Management and Metro Man

- His projects complete well before target date
- Major Projects
  - Restoring washed away Pamban Railway bridge in just 46 days instead of 6 month deadline set by railway (1964)
  - Kolkata Metro (1970)
  - Konkan Railway (1998)
  - Delhi Metro (1998-2011)
  - Kochi Metro



[Lessons from metro chief E Sreedharan's life for CEOs, managers & policymakers - The Economic Times (indiatimes.com)]

Thank You